

APOLLO INSTITUTE OF COMPUTER APPLICATION

BACHELOR OF COMPUTER APPLICATION

SEMESTER 03

Design Thinking and Innovation (BC03001071)

Lecture Notes

Unit 2: Empathy and User Research (6 Hours)

Q1. What is Empathy? Why is Empathy considered the foundation of Design Thinking? (GTU Summer 2026)

Definition

Empathy is the process of understanding users by observing, listening, interacting, and experiencing their feelings and challenges to design solutions that truly meet their needs.

Why is Empathy Considered the Foundation of Design Thinking?

Empathy is called the **foundation of Design Thinking** because all other stages of Design Thinking depend on understanding the user correctly.

Without empathy:

- Designers make assumptions instead of collecting real information.
- The actual problem may remain unidentified.
- The final product may fail to satisfy users.

With empathy:

- Real user needs are identified.
- Problems are defined accurately.
- Better ideas are generated.
- More useful prototypes are developed.
- Final solutions become user-centered.

Therefore, empathy serves as the base upon which the entire Design Thinking process is built.

Importance of Empathy

Empathy provides several benefits during the design process.

1. Understands Real User Needs

Empathy helps identify what users actually need rather than what designers think they need.

2. Identifies Hidden Problems

Many users cannot clearly explain their difficulties. Through observation and interaction, designers can discover problems that users themselves may not notice.

3. Improves User Experience

Products developed with empathy are easier to understand, comfortable to use, and provide a better experience.

4. Encourages Innovation

When designers understand users deeply, they can create creative and innovative solutions for real-world problems.

5. Reduces Product Failure

Products designed according to user needs are more likely to be accepted in the market, reducing the chances of failure.

6. Builds Customer Satisfaction

When users feel that a product solves their problems effectively, customer satisfaction and trust increase.

7. Supports Better Decision Making

Empathy provides reliable user information, enabling designers to make informed design decisions.

Q2. Explain "Know Your User" in detail. (GTU Winter 2025)

Definition

Know Your User is the process of collecting and analyzing information about users to understand their needs, goals, behaviors, emotions, expectations, and challenges before designing a solution.

It is the first step in the **Empathy Stage** of the Design Thinking process.

Why is "Know Your User" Important?

Understanding users helps designers create products that are useful, easy to use, and meaningful.

Without understanding users:

- Wrong problems may be solved.
- Designers make assumptions instead of using facts.
- Products may fail in the market.
- User satisfaction decreases.

With proper user understanding:

- Real problems are identified.
 - Better ideas are generated.
 - Products become more user-friendly.
 - Customer satisfaction increases.
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Objectives of Knowing Your User

The main objectives are:

1. To understand users' needs and expectations.
 2. To identify the problems faced by users.
 3. To study users' behavior and habits.
 4. To understand users' emotions and motivations.
 5. To create products that provide a better user experience.
 6. To reduce the chances of product failure.
 7. To develop innovative and user-centered solutions.
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Methods to Know Your User

Designers use different research methods to understand users.

1. Interview

An interview is a direct conversation with users to understand their experiences, opinions, and expectations.

2. Observation

Observation involves watching users while they perform tasks in real-life situations.

3. Surveys and Questionnaires

A survey collects information from many users using structured questions.

4. User Feedback

Designers collect suggestions, complaints, and reviews from users after they use a product.

5. Persona Creation

A persona is a fictional profile representing a typical user based on collected research data.

6. Role Playing

Designers act like users to experience their problems and understand their emotions.

Process of "Know Your User"

Identify Target Users



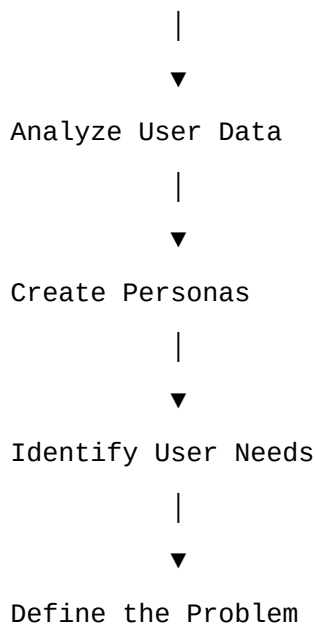
Collect User Information



Conduct Interviews



Observe Users



Real-Life Example: Smart Classroom

A college plans to develop a smart classroom system.

After conducting interviews and observing students, designers find:

Problems

- Students miss assignment deadlines.
- Lecture materials are scattered across different platforms.
- Important announcements are difficult to find.

Solution

The college develops a single portal that includes:

- Assignment reminders
- Lecture notes
- Attendance records
- Timetable
- Notifications
- Online submissions

Students can now access all academic information from one place.

Advantages of Knowing Your User

1. Helps understand real customer needs.

2. Improves product quality.
 3. Reduces design mistakes.
 4. Encourages innovation.
 5. Increases customer satisfaction.
 6. Supports better decision-making.
 7. Reduces development cost.
 8. Increases product success.
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Limitations

1. Time-consuming process.
2. Requires skilled researchers.
3. Users may provide incomplete information.
4. Research can be expensive.
5. Different users may have different expectations

Q3. Explain User Research. Discuss different User Research Methods.

(Based on syllabus)

Definition of User Research

User Research is the systematic process of collecting, analyzing, and understanding information about users, their needs, behaviors, goals, expectations, and challenges to design user-centered products and services.

In simple words, User Research means studying users to understand their problems before creating a solution.

Types of User Research

User Research is mainly classified into **two types**.

1. Primary User Research

Primary research involves collecting **new and original data directly from users**.

Examples

- Interviews
- Observation
- Surveys
- Focus Groups

- Role Playing

Advantages

- Accurate and reliable information
- Based on real user experiences
- Helps identify current problems

Disadvantages

- Time-consuming
 - Costly
 - Requires planning
-

2. Secondary User Research

Secondary research involves collecting information from **existing sources**.

Sources

- Books
- Research papers
- Websites
- Government reports
- Company reports
- Market research reports

Advantages

- Less expensive
- Saves time
- Easy to access

Disadvantages

- Information may be outdated.
 - Data may not match the current problem.
 - Less accurate than primary research.
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Different User Research Methods

Designers use different methods to understand users.

1. Interview

Definition

An **Interview** is a direct conversation with users to understand their experiences, opinions, expectations, and problems.

Example

A college interviews students to know the difficulties they face while using an online learning portal.

Advantages

- Detailed information
- Better understanding of user needs
- Opportunity to ask follow-up questions

Disadvantages

- Time-consuming
 - Requires good communication skills
-

2. Observation

Definition

Observation is the process of watching users while they perform their activities to understand their actual behavior.

Example

Observing customers while using an ATM to identify usability problems.

Advantages

- Reveals actual user behavior
- Identifies hidden problems
- Provides realistic information

Disadvantages

- Time-consuming
 - Privacy concerns
 - Observer bias
-

3. Surveys and Questionnaires

Definition

A survey is a research method used to collect information from many users by asking a set of structured questions.

Example

A company sends an online questionnaire to customers to collect feedback about a shopping application.

Advantages

- Collects information from many users quickly
- Cost-effective
- Easy to analyze

Disadvantages

- Users may not answer honestly.
 - Limited detailed responses.
-

4. Persona Creation

Definition

A **Persona** is a fictional representation of a typical user created using real research data.

It helps designers understand users' needs and behavior.

Example

Name: Rahul Patel

Age: 20 years

Occupation: BCA Student

Goal: Submit assignments on time

Problem: Misses deadlines due to multiple notifications.

5. Role Playing

Definition

Role Playing is a technique in which designers act like users to understand their experiences and emotions.

Example

Designers use wheelchairs to experience the challenges faced by physically disabled users.

Advantages

- Builds empathy
 - Identifies hidden problems
 - Improves creativity
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6. User Feedback

Definition

User Feedback is the process of collecting users' opinions after they use a product or service.

Sources of Feedback

- Customer reviews
- Ratings
- Complaint forms
- Social media comments
- Feedback forms

Example

An online shopping company improves its mobile application after reading customer reviews.

Q4. Explain Interview Method in User Research.

Definition

An Interview is a user research method in which a researcher directly interacts with users by asking planned questions to understand their needs, opinions, experiences, behaviors, and problems.

In simple words, **an interview is a face-to-face or online conversation with users to gather detailed information.**

Objectives of Interview Method

The main objectives of conducting interviews are:

1. To understand users' needs and expectations.
2. To identify users' problems and pain points.
3. To understand users' emotions and motivations.
4. To gather detailed qualitative information.
5. To validate assumptions made by designers.
6. To improve product and service design.
7. To support better decision-making.

Types of Interviews

1. Structured Interview

- Uses a fixed list of questions.
- Every participant is asked the same questions.
- Easy to compare responses.

Example

A researcher asks all students the same questions about the college website.

2. Semi-Structured Interview

- Contains prepared questions but allows additional questions.
- Offers flexibility during the discussion.
- Most commonly used in Design Thinking.

Example

A researcher asks students about online learning and asks extra questions based on their responses.

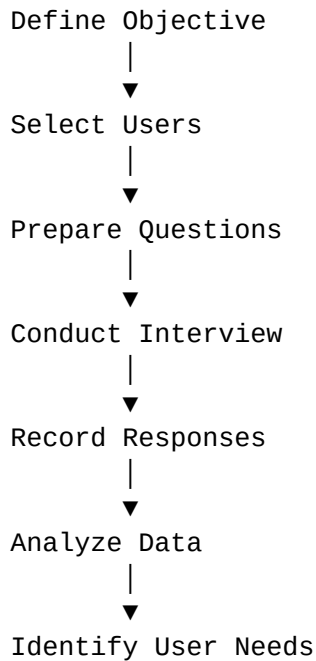
3. Unstructured Interview

- No fixed questions.
- Conversation flows naturally.
- Used to gain deep insights into user experiences.

Example

Talking freely with elderly people to understand their experience with digital banking.

Interview Process Flowchart



Examples of Interview Questions

Open-ended Questions

These encourage users to give detailed answers.

Examples:

- What difficulties do you face while using online banking?
- How do you usually order food online?
- What features would you like to add to this application?

Closed-ended Questions

These require short answers.

Examples:

- Do you use online payment? (Yes/No)

- Have you used this application before? (Yes/No)
-

Real-Life Example

Problem

A university wants to improve its **Online Learning Portal**.

Interview Conducted

The university interviews students and faculty members.

Questions Asked

- What problems do you face while using the portal?
- Which features are difficult to use?
- What improvements do you suggest?

Findings

- Students miss assignment deadlines.
- Navigation is confusing.
- Notifications are not clear.
- Lecture materials are difficult to locate.

Solution

The university redesigns the portal by adding:

- Assignment reminders
- Improved navigation
- Search facility
- Personalized dashboard
- Better notification system

As a result, students can easily access learning resources and submit assignments on time.

Q5. What is Observation? Explain its types. (GTU Winter 2025)

Definition

Observation is a User Research method in which designers carefully watch users while they perform their activities to understand their behavior, needs, problems, and interactions with a product or service.

In simple words, **Observation means learning about users by watching them perform real tasks.**

Types of Observation

Observation can be classified into four main types.

1. Direct Observation

Definition

In **Direct Observation**, the researcher personally watches users while they perform their activities.

The researcher does not interfere but carefully records users' actions and behavior.

Example

A designer observes customers using an ATM to identify difficulties in withdrawing money.

Advantages

- Provides first-hand information.
 - Easy to identify user behavior.
 - Helps discover hidden problems.
-

2. Indirect Observation

Definition

In **Indirect Observation**, the researcher observes users through recordings or monitoring systems instead of being physically present.

Example

Studying CCTV footage to analyze customer movement inside a shopping mall.

Advantages

- Users behave naturally.
 - Observation can be repeated.
 - Suitable for large areas.
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3. Participant Observation

Definition

In **Participant Observation**, the researcher actively participates in the users' activities to experience their situation personally.

This helps the researcher develop empathy.

Example

A designer travels in a crowded city bus to understand the difficulties faced by daily commuters.

Another example is using a wheelchair to understand accessibility problems faced by physically disabled people.

Advantages

- Better understanding of users.
 - Develops empathy.
 - Identifies hidden problems.
 - Provides real experience.
-

4. Non-Participant Observation

Definition

In **Non-Participant Observation**, the researcher only watches users without becoming part of the activity.

Example

Watching students use a library management system without interacting with them.

Advantages

- Users behave naturally.
 - Less influence on users' behavior.
 - Easy to record observations.
-

Observation Process

The observation method follows these steps.

Step 1: Define the Objective

Clearly identify what needs to be observed.

Example:

Observe how customers use a self-service kiosk.

Step 2: Select Users

Choose the appropriate target users.

Example:

Students, customers, patients, teachers, etc.

Step 3: Observe Users

Watch users carefully while they perform their activities.

Do not interrupt or influence them.

Step 4: Record Observations

Record important information using:

- Notes
 - Photographs
 - Videos
 - Audio recordings
-

Step 5: Analyze Data

Identify:

- User behavior
 - Common mistakes
 - User frustrations
 - Improvement opportunities
-

Step 6: Suggest Solutions

Use the findings to improve the product or service.

Real-Life Example

Problem

A hospital receives complaints about long waiting times.

Observation Conducted

The design team observes patients during the registration process.

Findings

- Long queues at registration.
- Patients repeatedly filling the same information.
- Confusing directions inside the hospital.
- Delay in token generation.

Solution

The hospital introduces:

- Online registration.
- Digital token system.
- Self-service kiosks.
- Directional signboards.

As a result, waiting time decreases and patient satisfaction improves.

Q6. What is Persona? Explain Persona Creation in detail. *(Asked in Winter 2025 and Summer 2026)*

Definition of Persona

A Persona is a fictional but realistic representation of a typical user, created using information collected through user research. It represents the characteristics, needs, goals, behaviors, and problems of a group of similar users.

In simple words,

A Persona is an imaginary user profile created from real user data to help designers understand their target users better.

Need for Persona

Creating a Persona helps designers:

- Understand users more effectively.
 - Focus on users instead of assumptions.
 - Design products according to user needs.
 - Improve communication within the design team.
 - Make better design decisions.
 - Build user-centered products.
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Objectives of Persona Creation

The objectives of creating a Persona are:

1. To understand users' needs and expectations.
 2. To represent a group of similar users.
 3. To identify users' goals and challenges.
 4. To improve product usability.
 5. To support user-centered design.
 6. To reduce design mistakes.
 7. To improve customer satisfaction.
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Characteristics of a Good Persona

A good Persona should:

- Be based on real user research.
 - Represent a specific user group.
 - Include realistic information.
 - Focus on users' goals and problems.
 - Be easy to understand.
 - Help guide design decisions.
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Components of a Persona

A Persona generally includes the following details:

Component	Description
Name	Fictional name of the user
Photograph	Representative image (optional)
Age	User's age
Gender	Male/Female/Other
Occupation	Profession or job

Component	Description
Education	Educational qualification
Location	City or region
Goals	What the user wants to achieve
Needs	User requirements
Challenges (Pain Points)	Problems faced by the user
Interests	Hobbies or preferences
Technology Usage	Devices and applications used
Motivation	Reasons for using the product

Steps of Persona Creation

Persona creation is a systematic process consisting of the following steps.

Step 1: Conduct User Research

Collect information about users through:

- Interviews
 - Observation
 - Surveys
 - Questionnaires
 - User Feedback
-

Step 2: Analyze the Data

Study the collected information carefully.

Identify:

- Similar behaviors
 - Common goals
 - Shared problems
 - User expectations
-

Step 3: Group Similar Users

Users with similar characteristics are grouped together.

Example:

- College Students
- Teachers

- Working Professionals
 - Senior Citizens
-

Step 4: Create the Persona

Prepare a fictional profile representing the group.

Include:

- Name
 - Age
 - Occupation
 - Goals
 - Needs
 - Problems
 - Technology usage
 - Motivations
-

Step 5: Validate the Persona

Compare the Persona with actual user research.

Ensure that it accurately represents the target users.

Step 6: Use Persona During Design

Use the Persona while:

- Defining the problem
 - Generating ideas
 - Developing prototypes
 - Testing solutions
-

Persona Creation Process

Conduct User Research

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Collect User Data

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Analyze User Behaviour



Group Similar Users



Create Persona Profile



Validate Persona



Use Persona in Design Process

Sample Persona

User Persona

Name: Rahul Patel

Age: 20 Years

Occupation: BCA Student

Education: Undergraduate

Location: Ahmedabad

Technology Used:

- Android Smartphone
- Laptop
- Google Classroom
- WhatsApp

Goals

- Submit assignments on time.
- Score good marks.
- Access study materials easily.

Needs

- Simple learning platform.

- Assignment reminders.
- Easy navigation.

Pain Points (Challenges)

- Too many WhatsApp notifications.
- Difficulty finding lecture notes.
- Misses assignment deadlines.

Motivation

- Wants to improve academic performance.
-

Real-Life Example

Problem

A company wants to develop an online grocery shopping application.

After conducting interviews and surveys, they create the following Persona.

Persona

Name: Neha Shah

Age: 35 Years

Occupation: Working Professional

Goals

- Buy groceries quickly.
- Save time.

Needs

- Fast delivery.
- Easy payment.
- Product availability.

Problems

- Long checkout process.
- Delayed deliveries.
- Difficult navigation.

Solution

The company improves the application by adding:

- One-click checkout

- Smart search
- Live delivery tracking
- Multiple payment options

As a result, customer satisfaction increases.

Advantages of Persona Creation

1. Helps understand users better.
 2. Supports user-centered design.
 3. Improves communication within the design team.
 4. Helps identify user needs and problems.
 5. Improves product quality.
 6. Saves development time.
 7. Reduces product failure.
 8. Increases customer satisfaction.
-

Limitations of Persona Creation

1. Depends on accurate user research.
 2. Time-consuming process.
 3. User needs may change over time.
 4. One Persona cannot represent every user.
 5. Poor research results in inaccurate Personas.
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Q7. Explain Role Playing in detail. (GTU Winter 2025)

Definition

Role Playing is a User Research technique in which designers or researchers act as users and perform real-life activities to understand users' experiences, emotions, needs, and challenges.

In simple words:

Role Playing means putting yourself in the user's place to experience their problems and understand their needs.

Objectives of Role Playing

The main objectives of Role Playing are:

1. To develop empathy towards users.

2. To understand users' emotions and experiences.
 3. To identify hidden problems that users may not express.
 4. To improve user experience.
 5. To generate innovative solutions.
 6. To validate design ideas.
 7. To create user-centereded products and services.
-

Role Playing Process

The Role Playing method follows a systematic process.

Step 1: Identify the User

Select the type of user whose experience needs to be understood.

Example:

Students, elderly people, patients, wheelchair users, farmers, etc.

Step 2: Define the Scenario

Decide the situation or activity to be performed.

Example:

Using a hospital registration system.

Step 3: Assign Roles

Participants are assigned different roles.

Examples:

- Customer
 - Student
 - Teacher
 - Shopkeeper
 - Patient
 - Bank Customer
-

Step 4: Perform the Activity

Participants perform the task exactly as real users would.

During this stage, researchers carefully observe:

- Difficulties faced

- Emotions
- User behavior
- Time taken
- Mistakes made

Step 5: Record Observations

Researchers record:

- User reactions
- Pain points
- Suggestions
- Opportunities for improvement

Step 6: Analyze Findings

The collected observations are analyzed to identify:

- User needs
- Problems
- Design improvements

Step 7: Develop Better Solutions

The final solution is modified based on the insights gained through Role Playing.

Role Playing Process Flowchart

Identify User

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Define Scenario

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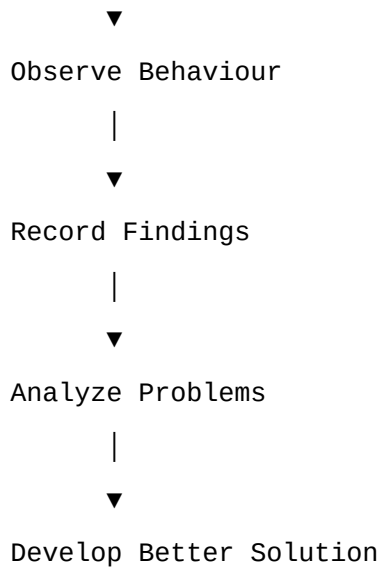
Assign Roles

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Perform Activity

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Applications of Role Playing

Role Playing is widely used in:

- Product Design
- Mobile App Development
- Website Design
- Healthcare Services
- Banking Systems
- Educational Applications
- Public Transportation
- Customer Service Improvement

Real-Life Example 1: Wheelchair Accessibility

Problem

A college wants to improve accessibility for physically disabled students.

Role Playing Activity

Designers use wheelchairs and move around the college campus.

Problems Observed

- Steep ramps
- Narrow doors
- Broken pathways

- High staircases
- Inaccessible washrooms

Solution

The college improves accessibility by:

- Building wider ramps
- Installing lifts
- Creating accessible washrooms
- Providing reserved parking

The campus becomes more accessible and user-friendly.

Q8. Differentiate Interview and Observation.

Basis of Comparison	Interview	Observation
Definition	Interview is a method of collecting information by asking questions directly to users.	Observation is a method of collecting information by watching users perform their activities.
Communication	Direct communication takes place between the researcher and the user.	No direct communication is required while observing users.
Data Collection	Data is collected through questions and answers.	Data is collected by observing users' behavior and actions.
Focus	Focuses on users' opinions, feelings, experiences, and expectations.	Focuses on users' actual behavior, actions, and interactions.
User Participation	Users actively participate by answering questions.	Users naturally perform their tasks while being observed.
Flexibility	Follow-up questions can be asked to get detailed information.	Follow-up questions cannot be asked during observation.
Accuracy	Depends on users' responses, which may sometimes be biased or incomplete.	Provides more realistic information because actual behavior is observed.
Time Required	Less time is required for individual interviews, but many interviews can be time-consuming.	Observation usually requires more time because users are watched during real activities.
Best Used For	Understanding users' thoughts, opinions, emotions, and expectations.	Understanding users' actual behavior and identifying hidden problems.
Example	Asking students about problems faced while using an online learning portal.	Watching students use the online learning portal to identify navigation problems.